

Calculus II Lesson 11

$$1) \sum_{n=0}^{\infty} 2^n = 2^0 + 2^1 + 2^2 + 2^3 + \cdots$$

Note: $a_n = 2^n$

Find the following partial sums.

a) $S_1 = a_1 = \underline{\quad ? \quad}$ b) $S_2 = a_1 + a_2 = \underline{\quad ? \quad}$

c) $S_3 = a_1 + a_2 + a_3 = \underline{\quad ? \quad}$ d) $S_4 = a_1 + a_2 + a_3 + a_4 = \underline{\quad ? \quad}$

$$2) \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^0 + \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots$$

$$\text{Note: } a_n = \left(\frac{1}{2}\right)^n$$

Find the following partial sums.

$$\text{a) } S_1 = a_1 = \underline{\quad ? \quad} \quad \text{b) } S_2 = a_1 + a_2 = \underline{\quad ? \quad}$$

$$\text{c) } S_3 = a_1 + a_2 + a_3 = \underline{\quad ? \quad} \quad \text{d) } S_4 = a_1 + a_2 + a_3 + a_4 = \underline{\quad ? \quad}$$

$$3) \sum_{n=0}^{\infty} \frac{n}{n+2} = \left(\frac{0}{0+2} \right) + \left(\frac{1}{1+2} \right) + \left(\frac{2}{2+2} \right) + \dots$$

Note: $a_n = \frac{n}{n+2}$

Find the following partial sums.

a) $S_1 = a_1 = \underline{\quad ? \quad}$ b) $S_2 = a_1 + a_2 = \underline{\quad ? \quad}$

c) $S_3 = a_1 + a_2 + a_3 = \underline{\quad ? \quad}$ d) $S_4 = a_1 + a_2 + a_3 + a_4 = \underline{\quad ? \quad}$

4) Telescoping Series $\sum_{n=1}^{\infty} \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$. Does the series converge or diverge?

$$\text{Note: } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n+2} - \frac{1}{n+3} \right) = 0$$

$$\text{Note: } a_1 = \frac{1}{3} - \frac{1}{4}; \quad a_2 = \frac{1}{4} - \frac{1}{5}; \quad a_3 = \frac{1}{5} - \frac{1}{6}$$

$$\text{a) } S_2 = \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) = \underline{\quad ? \quad}$$

$$\text{b) } S_3 = \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) = \underline{\quad ? \quad}$$

$$\text{c) } S_4 = \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) + \left(\frac{1}{6} - \frac{1}{7} \right) = \underline{\quad ? \quad}$$

$$\begin{aligned} \text{d) } S_{10} &= \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) + \left(\frac{1}{6} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{8} \right) + \left(\frac{1}{8} - \frac{1}{9} \right) \\ &\quad + \left(\frac{1}{9} - \frac{1}{10} \right) + \left(\frac{1}{10} - \frac{1}{11} \right) + \left(\frac{1}{11} - \frac{1}{12} \right) + \left(\frac{1}{12} - \frac{1}{13} \right) = \underline{\quad ? \quad} \end{aligned}$$

$$\begin{aligned} \text{e) } S_k &= \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{8}\right) + \cdots \\ &+ \left(\frac{1}{(k-1)+2} - \frac{1}{(k-1)+3}\right) + \left(\frac{1}{k+2} - \frac{1}{k+3}\right) = \underline{\hspace{1cm}} \text{ (multiple choice)} \end{aligned}$$

$$\text{A) } \frac{1}{3} - \frac{1}{k+1} \quad \text{B) } \frac{1}{3} + \frac{1}{k} \quad \text{C) } \frac{1}{3} - \frac{1}{k+3} \quad \text{D) } \frac{1}{3} - \frac{1}{k+5}$$

$$\text{f) } \sum_{n=1}^{\infty} \left(\frac{1}{n+2} - \frac{1}{n+3} \right) = \lim_{k \rightarrow \infty} S_k = \underline{\hspace{1cm}}$$

$$\text{g) Does the Telescoping Series } \sum_{n=1}^{\infty} \left(\frac{1}{n+2} - \frac{1}{n+3} \right) \text{ converge or diverge?}$$

5) Telescoping Series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 3n + 2}$. Does the series converge or diverge?

$$\text{Note: } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n^2 + 3n + 2} = 0$$

$$\text{Decomposing } \frac{1}{n^2 + 3n + 2}: \quad \frac{1}{n^2 + 3n + 2} = \frac{1}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2} = \frac{1}{n+1} - \frac{1}{n+2}$$

$$\text{Note: } a_1 = \frac{1}{2} - \frac{1}{3}; \quad a_2 = \frac{1}{3} - \frac{1}{4}; \quad a_3 = \frac{1}{4} - \frac{1}{5}$$

$$\text{a) } S_2 = \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) = \underline{\quad ? \quad}$$

$$\text{b) } S_3 = \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) = \underline{\quad ? \quad}$$

$$\text{c) } S_4 = \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) = \underline{\quad ? \quad}$$

$$\text{d) } S_{10} = \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) + \left(\frac{1}{6} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{8} \right)$$

$$+ \left(\frac{1}{8} - \frac{1}{9} \right) + \left(\frac{1}{9} - \frac{1}{10} \right) + \left(\frac{1}{10} - \frac{1}{11} \right) + \left(\frac{1}{11} - \frac{1}{12} \right) = \underline{\quad ? \quad}$$

$$\text{e) } S_k = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \dots$$

$$+ \left(\frac{1}{(k-1)+1} - \frac{1}{(k-1)+2}\right) + \left(\frac{1}{k+1} - \frac{1}{k+2}\right) = \underline{\hspace{1cm}} \text{ (multiple choice)}$$

A) $\frac{1}{2} - \frac{1}{k+1}$ B) $\frac{1}{2} + \frac{1}{k}$ C) $\frac{1}{2} - \frac{1}{k+2}$ D) $\frac{1}{4} - \frac{1}{k+3}$

$$\text{f) } \sum_{n=1}^{\infty} \frac{1}{n+1} - \frac{1}{n+2} = \lim_{k \rightarrow \infty} S_k = \underline{\hspace{1cm}}$$

g) Does the Telescoping Series $\sum_{n=1}^{\infty} \frac{1}{n+1} - \frac{1}{n+2}$ converge or diverge?

6) Telescoping Series $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$. Does the series converge or diverge?

Note: $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \ln\left(\frac{n+1}{n}\right) = \ln(1) = 0$

$$\ln\left(\frac{n+1}{n}\right) = \ln(n+1) - \ln(n)$$

Note: $a_1 = \ln(2) - \ln(1)$; $a_2 = \ln(3) - \ln(2)$; $a_3 = \ln(4) - \ln(3)$

a) $S_2 = [\ln(2) - \ln(1)] + [\ln(3) - \ln(2)] = \underline{\quad ? \quad}$

b) $S_3 = [\ln(2) - \ln(1)] + [\ln(3) - \ln(2)] + [\ln(4) - \ln(3)] = \underline{\quad ? \quad}$

c) $S_4 = [\ln(2) - \ln(1)] + [\ln(3) - \ln(2)] + [\ln(4) - \ln(3)] + [\ln(5) - \ln(4)] = \underline{\quad ? \quad}$

d) $S_7 = [\ln(2) - \ln(1)] + [\ln(3) - \ln(2)] + [\ln(4) - \ln(3)] + [\ln(5) - \ln(4)] + [\ln(6) - \ln(5)]$
 $+ [\ln(7) - \ln(6)] + [\ln(8) - \ln(7)] = \underline{\quad ? \quad}$

e) $S_k = [\ln(2) - \ln(1)] + [\ln(3) - \ln(2)] + [\ln(4) - \ln(3)] + [\ln(5) - \ln(4)] + [\ln(6) - \ln(5)] + \dots$
 $+ [\ln((k-1)+1) - \ln(k-1)] + [\ln(k+1) - \ln(k)] = \underline{\hspace{2cm}}$ (multiple choice)

A) $-\ln(1)+\ln(k)$ B) $-\ln(1)+\ln(k+1)$ C) $-\ln(1)+\ln(k+2)$ D) $-\ln(1)+\ln(k+3)$

f) $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right) = \lim_{k \rightarrow \infty} S_k = \underline{\hspace{2cm}}$

g) Does the Telescoping Series $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$ converge or diverge?

7) $\sum_{n=1}^{\infty} \frac{4}{n(n+1)}$ Note: $\sum_{n=1}^{\infty} \frac{4}{n(n+1)}$ is a telescoping series.

Using partial fraction decomposition to decompose $\frac{4}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1} = \frac{1}{n} + \frac{-1}{n+1} = \frac{1}{n} - \frac{1}{n+1}$

$$\sum_{n=1}^{\infty} \frac{4}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

Note: $a_1 = \frac{1}{1} - \frac{1}{2}$; $a_2 = \frac{1}{2} - \frac{1}{3}$; $a_3 = \frac{1}{3} - \frac{1}{4}$

a) $S_2 = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) = \underline{\hspace{2cm}}$

b) $S_3 = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) = \underline{\hspace{2cm}}$

c) $S_4 = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) = \underline{\hspace{2cm}}$

d) $S_{10} = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) + \left(\frac{1}{6} - \frac{1}{7} \right)$

$$+ \left(\frac{1}{7} - \frac{1}{8} \right) + \left(\frac{1}{8} - \frac{1}{9} \right) + \left(\frac{1}{9} - \frac{1}{10} \right) + \left(\frac{1}{10} - \frac{1}{11} \right) = \underline{\hspace{2cm}}$$

$$\text{e) } S_k = \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \dots$$

$$+ \left(\frac{1}{k-1} - \frac{1}{(k-1)+1}\right) + \left(\frac{1}{k} - \frac{1}{k+1}\right) = \underline{\hspace{1cm}} \text{ (Multiple Choice)}$$

A) $1 - \frac{1}{k+2}$ B) $1 - \frac{1}{k}$ C) $1 - \frac{1}{k+1}$ D) $2 - \frac{1}{k}$

$$\text{f) } \sum_{n=1}^{\infty} \frac{4}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right) = \lim_{k \rightarrow \infty} S_k = \underline{\hspace{1cm}} \quad \text{(numerical value)}$$

8) Find the infinite series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 8n + 7} = \sum_{n=1}^{\infty} \left(\frac{1/6}{n+1} - \frac{1/6}{n+7} \right) = \frac{1}{6} \sum_{n=1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+7} \right)$

Let $S_k = \sum_{n=1}^k \left(\frac{1}{n+1} - \frac{1}{n+7} \right)$ sum of first k terms

$$\begin{aligned} S_k = & \left(\frac{1}{2} - \frac{1}{8} \right) + \left(\frac{1}{3} - \frac{1}{9} \right) + \left(\frac{1}{4} - \frac{1}{10} \right) + \left(\frac{1}{5} - \frac{1}{11} \right) + \left(\frac{1}{6} - \frac{1}{12} \right) + \left(\frac{1}{7} - \frac{1}{13} \right) + \left(\frac{1}{8} - \frac{1}{14} \right) \\ & + \left(\frac{1}{9} - \frac{1}{15} \right) + \left(\frac{1}{10} - \frac{1}{16} \right) + \left(\frac{1}{11} - \frac{1}{17} \right) + \left(\frac{1}{12} - \frac{1}{18} \right) + \left(\frac{1}{13} - \frac{1}{19} \right) + \left(\frac{1}{14} - \frac{1}{20} \right) \\ & + \dots + \left(\frac{1}{k-7} - \frac{1}{k-3} \right) + \left(\frac{1}{k-6} - \frac{1}{k-2} \right) + \left(\frac{1}{k-5} - \frac{1}{k-1} \right) + \left(\frac{1}{k-8} - \frac{1}{k-2} \right) \\ & + \left(\frac{1}{k-7} - \frac{1}{k-1} \right) + \left(\frac{1}{k-6} - \frac{1}{k} \right) + \left(\frac{1}{k-5} - \frac{1}{k+1} \right) + \left(\frac{1}{k-4} - \frac{1}{k+2} \right) \\ & + \left(\frac{1}{k-3} - \frac{1}{k+3} \right) + \left(\frac{1}{k-2} - \frac{1}{k+4} \right) + \left(\frac{1}{k-1} - \frac{1}{k+5} \right) + \left(\frac{1}{k} - \frac{1}{k+6} \right) \\ & + \left(\frac{1}{k+1} - \frac{1}{k+7} \right) \end{aligned}$$

$$S_k = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \dots - \frac{1}{k+2} - \frac{1}{k+3} - \frac{1}{k+4} - \frac{1}{k+5} - \frac{1}{k+6} - \frac{1}{k+7}$$

a) $\lim_{k \rightarrow \infty} =$ _____ ?

b) $\sum_{n=1}^{\infty} \frac{1}{n^2 + 8n + 7} = \frac{1}{6} \sum_{n=1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+7} \right) =$ _____ ?

